<<Partial Differential equation>> . Evans . 习题

Chapter 2: 1. $\begin{cases} u_t + b\cdot Du + cu = 0 & \text{in } \mathbb{R}^n \times (0,\infty) \\ u = g & \text{on } \mathbb{R}^n \times \{t=0\} \end{cases}$ $\quad c\in\mathbb{R}$ and $b\in\mathbb{R}^n$ are constant.

Solution: $z(s) := u(x+sb, t+s)$, $\dot{z}(s) = Du(x+sb, t+s)\cdot b + u_t(x+sb, t+s) = -cu(x+sb, t+s) = -cz(s)$

$$\Rightarrow \frac{z'}{z} = -c \Rightarrow (\ln z)' = -c \Rightarrow \ln z(s) = -cs + c_1 \Rightarrow z(s) = e^{-cs+c_1} = c_1 e^{-cs} \quad (c_1\text{与}s\text{无关})$$

$$z(-t) = u(x-tb, 0) = g(x-tb) = c_1 e^{ct} \Rightarrow c_1 = e^{-ct} g(x-tb) \quad , \quad u(x,t) = z(0) = c_1 = e^{-ct} g(x-tb)$$

2. Prove $\Delta = 0$, if O is an orthogonal $n\times n$ matrix the $v(x) = u(Ox)$, $\Delta v(x) = 0$

Solution : $O = (O_{ij})_{n\times n}$, $y_i = \sum_{j=1}^n O_{ij}x_j$, Then $v(x) = v(x_1 \cdots x_n) = u(Ox) = u(y_1 \cdots y_n)$

$y_1 = O_{11}x_1 + O_{12}x_2 + \cdots O_{1n}x_n$, $\quad v_{x_1} = u_{y_1}O_{11} + u_{y_2}O_{21} + \cdots u_{y_n}O_{n1} \qquad v_{x_k} = \sum_{i=1}^n u_{y_i}O_{ik}$

$y_2 = O_{21}x_1 + O_{22}x_2 + \cdots O_{2n}x_n \qquad v_{x_kx_k} = \sum_{i=1}^n\sum_{j=1}^n u_{y_iy_j}O_{ik}O_{jk}$

$\vdots$

$y_n = O_{n1}x_1 + \cdots \quad O_{nn}x_n$

$$\Delta v = \sum_{k=1}^n v_{x_kx_k} = \sum_{k=1}^n\sum_{i=1}^n\sum_{j=1}^n u_{y_iy_j}O_{ik}O_{jk} = \sum_{i=1}^n\sum_{j=1}^n \Big(u_{y_iy_j}\sum_{k=1}^n O_{ik}O_{jk}\Big) = \sum_{i=1}^n u_{y_iy_i} = 0$$

3. Show $n\geq 3$ $\quad u(0) = \fint_{\partial B(0,r)} g\,ds + \frac{1}{n(n-2)\alpha(n)}\int_{B(0,r)} \left(\frac{1}{|x|^{n-2}} - \frac{1}{r^{n-2}}\right) f\,dx$

$\begin{cases} -\Delta u = f & \text{in } B(0,r) \\ u = g & \text{on } \partial B(0,r) \end{cases}$

Solution $\quad \phi(s) := \fint_{\partial B(0,s)} u(x)\,ds = \fint_{\partial B(0,1)} u(sy)\,ds(y), \qquad \phi(r) = \fint_{\partial B(0,r)} g\,ds$

$\lim_{s\to 0}\phi(s) = u(0)$, $\quad \phi'(s) = \fint_{\partial B(0,1)} Du(sy)\cdot y\,ds(y) = \fint_{\partial B(0,s)} Du(x)\cdot\frac{x}{s}\,ds(x)$

$$= \fint_{\partial B(0,s)} \frac{\partial u}{\partial \nu}\,ds = \frac{s}{n}\fint_{B(0,s)} \Delta u(x)\,dx = -\frac{s}{n}\fint_{B(0,s)} f\,dx$$

$$= -\frac{1}{n\alpha(n)s^{n-1}}\int_{B(0,s)} f\,dx$$

$$\phi(r) - u(0) = \int_0^r \phi'(r)\,ds = -\frac{1}{n\alpha(n)}\int_0^r \Big(s^{1-n}\int_{B(0,s)} f\,dx\Big)\,ds$$

4. if $u\in C^2(U)\cap C(\bar{U})$ is harmonic , then $\max_{\bar{U}} u = \max_{\partial U} u$

Proof: Assume $\exists\ x_0\in U$ $\ \max_{\bar{U}} u_\varepsilon(x) = u_\varepsilon(x_0)$, then $Du_\varepsilon(x_0) = 0$, $\ D^2u_\varepsilon(x_0) \leq 0$

$\Rightarrow \Delta u_\varepsilon(x_0) \leq 0$, but $\Delta u_\varepsilon = \Delta u + \varepsilon\Delta|x|^2 = 2n\varepsilon > 0$

5. We say $v \in C^2(\bar{U})$ is subharmonic if $-\Delta v \le 0$ in Ω

(a) prove $v(x) \le ⨍_{B(x,r)} v\,dy$

proof: $\phi(r) := ⨍_{\partial B(x,r)} v(y)\,ds = ⨍_{\partial B(0,1)} v(x+rz)\,ds$, $\phi'(r) = ⨍_{\partial B(0,1)} Dv(x+rz)\cdot z\,ds$

$= ⨍_{\partial B(x,r)} \frac{\partial u}{\partial \nu}\,ds = \frac{1}{n\alpha(n) r^{n-1}} \int_{B(x,r)} \Delta u(y)\,dy \geq 0$, $\phi(r) \geq v(x)$

(b) $\max_{\bar{U}} v = \max_{\partial U} v$

proof: $\exists x_0 \in U$, $v(x_0) = M := \max_{\bar{U}} v$, $0 < r < \operatorname{dist}(x_0, \partial U)$

$$M = v(x_0) \le ⨍_{B(x_0,r)} v\,dy \le M \Rightarrow ⨍_{B(x_0,r)} v\,dy = M$$

$v(y) = M$ all $y \in B(x_0, r)$ $\Rightarrow$ $v(x) = M$ all $x \in \bar{U}$

(c) $\phi: \mathbb{R} \to \mathbb{R}$ be smooth and convex, u is harmonic $v := \phi(u)$

then v is subharmonic

proof: $\frac{\partial v}{\partial x_i} = \frac{\partial \phi}{\partial u} \frac{\partial u}{\partial x_i}$, $\frac{\partial^2 v}{\partial x_i^2} = \frac{d^2\phi}{du^2}\left(\frac{\partial u}{\partial x_i}\right)^2 + \frac{d\phi}{du}\frac{\partial^2 u}{\partial x_i^2}$

$$-\Delta v = -\frac{d^2\phi}{du^2} \sum_{i=1}^{n} \left(\frac{\partial u}{\partial x_i}\right)^2 - 0 \le 0$$

(d) Prove $v := |Du|^2$ is subharmonic u is harmonic

$v = |u_{x_1}, u_{x_2} \cdots u_{x_n}|^2 = u_{x_1}^2 + \cdots u_{x_n}^2$, $\frac{\partial v}{\partial x_i} = \sum_{j=1}^{n} 2 \frac{\partial u}{\partial x_j} \frac{\partial^2 u}{\partial x_i \partial x_j}$

$\frac{\partial^2 v}{\partial x_i^2} = 2\sum_{j=1}^{n} \frac{\partial}{\partial x_i}\left(\frac{\partial u}{\partial x_j} \frac{\partial^2 u}{\partial x_i \partial x_j}\right) = 2\sum_{j=1}^{n}\left[\left(\frac{\partial^2 u}{\partial x_i x_j}\right)^2 + \frac{\partial u}{\partial x_j}\frac{\partial^3 u}{\partial x_i^2 \partial x_j}\right]$

$-\Delta v = -2\sum_{i=1}^{n}\sum_{j=1}^{n}\left[\left(\frac{\partial^2 u}{\partial x_i \partial x_j}\right)^2 + \frac{\partial u}{\partial x_j}\frac{\partial^3 u}{\partial x_i^2 \partial x_j}\right] \le 0$

6. U be bounded open set of $\mathbb{R}^n$. $\max_{\bar{U}} |u| \le C\left(\max_{\partial U} |g| + \max_{\bar{U}} |f|\right)$ $\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U \end{cases}$

proof: $-\Delta\left(\pm u + \frac{|x|^2}{2n}\lambda\right) = \mp \Delta u - \frac{\lambda}{2n}\Delta |x|^2 = \pm f - \lambda \le 0$ $\quad (\lambda = \max_{\bar{U}} |f|)$

$\max_{\bar{U}} \pm u \le \max_{\bar{U}} \left(\pm u + \frac{|x|^2}{2n}\lambda\right) = \max_{\partial} \left(\pm g + \frac{|x|^2}{2n}\lambda\right) \le \max_{\partial U} |g| + \frac{1}{2n}\max_{\partial U} |x|^2 \max_{\bar{U}} |f| \le C\left(\max |g| + \max_{\bar{U}} |f|\right)$

7. Prove $r^{n-2}\frac{r-|x|}{(r+|x|)^{n-1}}u(0)\le u(x)\le r^{n-2}\frac{r+|x|}{(r-|x|)^{n-1}}u(0)$ $[\,u>0,\ \Delta u=0 \text{ in } B(0,r)\,]$

Proof: $g(x)=u(x)$ on $\partial B(0,r)$ Then $u(x)=\frac{r^2-|x|^2}{n\alpha(n)r}\int_{\partial B(0,r)}\frac{g(y)}{|x-y|^n}dS(y)$ $(x\in B(0,r))$

$u(0)=⨍_{\partial B(0,r)} g(y)\,dS(y)$, $u(x)\ge\frac{r^2-|x|^2}{n\alpha(n)r}\int_{\partial B(0,r)}\frac{g(y)}{(r+|x|)^n}dS(y)$ $\quad |x-y|\le|x|+|y|\le|x|+r$

$=\frac{r-|x|}{n\alpha(n)r(r+|x|)^{n-1}}\int_{\partial B(0,r)}g(y)\,dS(y)=r^{n-2}\frac{r-|x|}{(r+|x|)^{n-1}}u(0)$

$u(x)\le\frac{r^2-|x|^2}{n\alpha(n)r}\int_{\partial B(0,r)}\frac{g(y)}{(r-|x|)^n}dS(y)=r^{n-2}\frac{r+|x|}{(r-|x|)^{n-1}}u(0)$

8. Prove Theorem 15 in §2.2.4 (Hint: $u\equiv1$ solves $g\equiv1$ $\int_{\partial B(0,1)}K(x,y)\,dS(y)=1$, $x\in B(0,1)$)

Theorem $g\in C(\partial B(0,r))$ define $u(x)=\frac{r^2-x^2}{n\alpha(n)r}\int_{\partial B(0,r)}\frac{g(y)}{|x-y|^n}dS(y)$ Then. (1) $u\in C^\infty(B^\circ(0,r))$

(2) $\Delta u=0$ in $B(0,r)$ (3) $\lim_{\substack{x\to x^0\\ x\in B^\circ(0,r)}}u(x)=g(x^0)$ for $x^0\in\partial B(0,r)$

(1) proof: $x\to K(x,y)$ is smooth for $x\ne y$, $u=\int_{\partial B(0,r)}K(x,y)\,g(y)\,dS(y)\in C^\infty(B^\circ(0,r))$

(2) proof: $x\to\frac{\partial G(x,y)}{\partial\nu}$ is harmoic $x\in B(0,1)$ $y\in\partial B(0,1)$ $\tilde u(x)=\int_{\partial B(0,1)}g(y)\frac{\partial G}{\partial\nu}dS(y)$ is harmonic

9. u be solution of $\begin{cases}\Delta u=0 & \text{in } \mathbb{R}^n_+\\ u=g & \text{on } \partial\mathbb{R}^n_+\end{cases}$ Assume g is bounded and $g(x)=|x|$ for $\partial\mathbb{R}^n_+$ $|x|\le1$ Show Du is not bounded near $x=0$ $\{x_n>0\}$

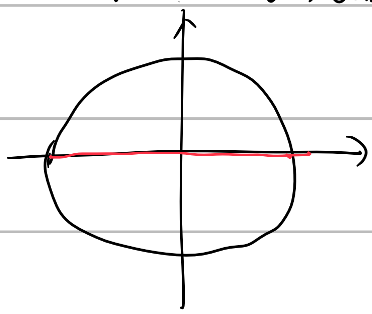
Proof: assume $n\ge2$, $u(x)=\frac{2x_n}{n\alpha(n)}\int_{\partial\mathbb{R}^n_+}\frac{g(y)}{|x-y|^n}dy$ $(x\in\mathbb{R}^n_+)$ $u(0)=g(0)=0$

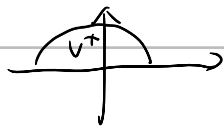
$u(\lambda e_n)=\frac{2\lambda}{n\alpha(n)}\int_{\partial\mathbb{R}^n_+}\frac{g(y)}{|\lambda e_n-y|^n}dy$ $\quad\frac{u(\lambda e_n)-u(0)}{\lambda}=\frac{2}{n\alpha(n)}\int_{\partial\mathbb{R}^n_+}\frac{g(y)}{|\lambda e_n-y|^n}dy$

$=\frac{2}{n\alpha(n)}\int_{\partial\mathbb{R}^n_+\cap B(0,1)}\frac{|y|}{|\lambda e_n-y|^n}dy+\frac{2}{n\alpha(n)}\int_{\partial\mathbb{R}^n_+\setminus B(0,1)}\frac{g(y)}{|\lambda e_n-y|^n}dy:=I+J$

$\lim_{\lambda\to0}I=\frac{2}{n\alpha(n)}\int_{\partial\mathbb{R}^n_+\cap B(0,1)}\lim_{\lambda\to0}\frac{|y|}{|\lambda e_n-y|^n}dy=\frac{2}{n\alpha(n)}\int_{\partial\mathbb{R}^n_+\cap B(0,1)}|y|^{1-n}dy=\frac{2}{n\alpha(n)}\int_0^1 r^{1-n}(n-1)\alpha(n-1)r^{n-2}dr$

$=\frac{2(n-1)\alpha(n-1)}{n\alpha(n)}\int_0^1 r^{-1}dr=\infty$





10. (Reflection principle) (a) $U^+=\{x\in\mathbb{R}^n \mid |x|<1, x_n>0\}$ Assume $u\in C^2(\overline{U^+})$ is harmonic

$u=0$ on $\partial U^+\cap\{x_n=0\}$ $\quad v(x)=\begin{cases} u(x) & x_n\geq 0 \\ -u(x_1\cdots x_{n-1},-x_n) & x_n<0\end{cases}$ Prove $v\in C^2(U)$ and harmonic

proof: $U^-=B(0,1)\setminus\overline{U^+}$ $\quad v(x)$在$\overline{U^+}$与U^-, C^2, $\quad w(x)=\frac{1-|x|^2}{n\alpha(n)}\int_{\partial U}\frac{v(y)}{|x-y|^n}dS(y)$

$\Rightarrow$ (i) $w\in C^\infty$ (ii) $\Delta w=0$ (iii) $w=v$ on ∂U

$x_n=0\Longrightarrow u=0\Rightarrow v=0\Rightarrow w=0 \;\Big|\; v-w=0$ on $\partial U^+ \;\Big|\Longrightarrow$ 在U^+ $u-w\overset{\text{极值原理}}{=}0$

同理 $w=v$ in $\overline{U^-}$ 故v在U内调和

11. kelvin transform for Laplace $Ku=\tilde{u}$ $\quad\tilde{u}=u(\bar{x})|\bar{x}|^{n-2}=u\left(\frac{x}{|x|^2}\right)|x|^{2-n}$

u is harmonic $\Longrightarrow$ so is $\tilde{u}$

12. suppose u is smooth. $u_t-\Delta u=0$ in $\mathbb{R}^n\times(0,\infty)$

(a) $u_\lambda(x,t)=u(\lambda x,\lambda^2 t)$ also solve. each $\lambda\in\mathbb{R}^n$ (b) $v(x,t)=x\cdot Du(x,t)+2tu_t(x,t)$ solve

Prove (a) $(u_\lambda)_t(x,t)=\lambda^2 u_t(\lambda x,\lambda^2 t)$ $\quad\Delta u_\lambda(x,t)=\lambda^2\Delta u(\lambda x,\lambda^2 t)\Longrightarrow (u_\lambda)_t-\Delta u_\lambda=0$

(b) $v_\lambda(x,t)=\frac{d}{d\lambda}u_\lambda(x,t)=x\cdot Du(\lambda x,\lambda^2 t)+2\lambda t\,u_t(\lambda x,\lambda^2 t)$

$(v_\lambda)_t(x,t)=\frac{d}{dt}\frac{d}{d\lambda}u_\lambda(x,t)=\frac{d}{d\lambda}(u_\lambda)_t(x,t)$ $\quad\Delta v_\lambda(x,t)=\frac{d}{d\lambda}\Delta u_\lambda(x,t)$

v_λ solve equation v_1 solve too.

13. Assume $n=1$ $u(x,t)=v\left(\frac{x}{\sqrt{t}}\right)$ (a) $u_t=u_{xx}\Leftrightarrow v''+\frac{z}{2}v'=0$, $v(z)=c\int_0^z e^{-\frac{s^2}{4}}ds+d$

proof: (a) $z=\frac{x}{\sqrt{t}}$ $\quad u_t=v'\left(-\frac{x}{2t\sqrt{t}}\right)=-\frac{z}{2t}v'$, $u_x=\frac{1}{\sqrt{t}}v'$, $u_{xx}=\frac{1}{t}v''$ $\quad u_t-u_{xx}=-\frac{1}{t}(v''+\frac{z}{2}v')=0$

(b) $u(x,t)=v\left(\frac{x}{\sqrt{t}}\right)=c\int_0^{\frac{x}{\sqrt{t}}}e^{-\frac{s^2}{4}}ds+d$ $\quad u_x=c\frac{1}{\sqrt{t}}e^{-\frac{x^2}{4t}}=\frac{1}{2\sqrt{\pi t}}e^{-\frac{x^2}{4t}}$.

15. Given $g:[0,\infty)\to\mathbb{R}$ $g(0)=0$, $u(x,t)=\frac{x}{\sqrt{4\pi}}\int_0^t\frac{1}{(t-s)^{\frac{3}{2}}}e^{\frac{-x^2}{4(t-s)}}g(s)ds$ for a solution of

$\begin{cases} u_t-u_{xx}=0 & \text{in } \mathbb{R}_+\times(0,\infty) \\ u=0 & \mathbb{R}_+\times\{t=0\} \\ u=g & \{x=0\}\times[0,\infty)\end{cases}$

21. $E=(E^1,E^2,E^3)$, $B=(B^1,B^2,B^3)$ solves Maxwell's equations $\begin{cases} E_t = \nabla\wedge B \\ B_t = -\nabla\wedge E \\ \nabla\cdot E = \nabla\cdot B = 0\end{cases} \Rightarrow \begin{matrix} E_{tt}-\Delta E=0 \\ B_{tt}-\Delta B = 0\end{matrix}$

proof: $E_t=\nabla\wedge B \Rightarrow E_{tt} = \nabla\wedge(B_t) = -\nabla\wedge\nabla\wedge E = \Delta E - \nabla(\nabla\cdot E)$

Chapter 5

1. $k=\{0,1,\dots\}$, $0<r\le 1$, Prove $C^{k,r}(\bar U)$ is Banach space

完备 + 赋范 = 完备 + 数乘 + 正定 + 三角.

① $\|u+v\|\le\|u\|+\|v\|$
$\|\lambda u\| = |\lambda|\,\|u\|$
$\|u\|=0 \Leftrightarrow u=0$

Proof: $\|u\|_{C^{k,r}(\bar U)} := \sum_{|\alpha|\le k}\|D^\alpha u\|_{C(\bar U)} + \sum_{|\alpha|=k}[D^\alpha u]_{C^r(\bar U)}$ $\quad [D^\alpha u]_{C^r} = \sup \frac{|D^\alpha u(x) - D^\alpha u(y)|}{|x-y|^r}$

$$[D^\alpha u + D^\alpha v] = \sup \frac{|D^\alpha(u(x)+v(x)) - (D^\alpha(u(y)+v(y)))|}{|x-y|^r} \le \sup\frac{|D^\alpha u(x)-D^\alpha u(y)|}{|x-y|^r} + \sup(\quad)$$

完备性, Cauchy sequence $\{u_k\}_{k=1}^\infty \subset C^{k,r}(\bar U)$, $\forall \varepsilon>0$, $\exists N>0$. $\|u_n-u_m\|_{C^{k,r}(\bar U)}$

$$= \sum_{|\alpha|\le k}\|D^\alpha(u_n-u_m)\|_{C(\bar U)} + \sum_{|\alpha|=k}[D^\alpha(u_n-u_m)]_{C^{0,r}(\bar U)} < \varepsilon$$

$C(\bar U)$ is Banach $\Rightarrow \exists\, u\in C^k(\bar U)$ $\quad u_n\to u$ in C^k $\quad [D^\alpha u_n(x) - D^\alpha u]_{C^r} = \frac{D^\alpha(u_n-u)(x) - D^\alpha(u_n-u)(y)}{|x-y|^r}\to 0$

2. $0<\beta<r\le 1$. Prove interpolation inequality $\|u\|_{C^{0,r}(U)} \le \|u\|_{C^{0,\beta}(U)}^{\frac{1-r}{1-\beta}}\|u\|_{C^{0,1}(U)}^{\frac{r-\beta}{1-\beta}}$

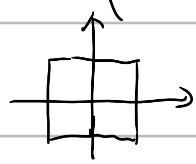
proof: $\|u\|_{C^{0,r}(U)} = \|u\|_{C(U)} + \sup_{\substack{x,y\in U\\ x\ne y}}\left\{\frac{|u(x)-u(y)|}{|x-y|^r}\right\}$ $\quad \beta\left(\frac{1-r}{1-\beta}\right)+\left(\frac{r-\beta}{1-\beta}\right) = r$

$\frac{1-r}{1-\beta}+\frac{r-\beta}{1-\beta} = \frac{1-\beta}{1-\beta}=1$ $\quad = \|u\|_{C(U)}^{\frac{1-r}{1-\beta}}\|u\|_{C(U)}^{\frac{r-\beta}{1-\beta}} + \sup_{\substack{x,y\in U\\ x\ne y}}\left\{\left(\frac{|u(x)-u(y)|}{|x-y|^\beta}\right)^{\frac{1-r}{1-\beta}}\left(\frac{|u(x)-u(y)|}{|x-y|}\right)^{\frac{r-\beta}{1-\beta}}\right\}$

$$\le \|u\|_{C(U)}^{\frac{1-r}{1-\beta}}\|u\|_{C(U)}^{\frac{r-\beta}{1-\beta}} + \sup\left\{\frac{|u(x)-u(y)|}{|x-y|^\beta}\right\}^{\frac{1-r}{1-\beta}} \cdot \sup\left\{\frac{|u(x)-u(y)|}{|x-y|}\right\}^{\frac{r-\beta}{1-\beta}}$$

$ab+cd\le(a+c)(b+d)$ $\quad \le \left(\|u\|_{C(U)}+\sup\left\{\frac{|u(x)-u(y)|}{|x-y|^\beta}\right\}\right)^{\frac{1-r}{1-\beta}}\left(\|u\|_{C(U)}+\sup\frac{|u(x)-u(y)|}{|x-y|}\right)^{\frac{r-\beta}{1-\beta}}$

$$=\|u\|_{C^{0,\beta}}^{\frac{1-r}{1-\beta}}\|u\|_{C^{0,1}}^{\frac{r-\beta}{1-\beta}}$$



3. Denote by U the open square $\{x\in\mathbb{R}^2 \mid |x_1|<1,\ |x_2|<1\}$, $u(x)=\begin{cases}1-x_1, & x_1>0 & |x_2|<x_1\\ 1+x_1, & x_1<0 & |x_2|<-x_1\\ 1-x_2, & x_2>0, & |x_1|<x_2 \\ 1+x_2, & x_2<0 & |x_1|<-x_1\end{cases}$

$1\le p\le\infty$. $u\in W^{1,p}(U)$?

$U_1=\{x_1>0, |x_2|<x_1\}$ $\quad U_2=\{x_1<0, |x_2|<-x_1\}$, $\quad U_3=\{x_2>0, |x_1|<x_2\}$ $\quad U_4=\{x_2<0, |x_1|<-x_2\}$

Solution. $D_1=\{0\le x_1=x_2\le 1\}$ $\quad D_2=\{-1\le x_1=-x_2\le 0\}$ $\quad D_3=\{-1\le x_1=x_2\le 0\}$ $\quad D_4=\{0\le x_1=-x_2\le 1\}$

$V_1(x) = \begin{cases} -1 & V_1 \\ 1 & V_2 \\ 0 & V_3 \\ 0 & V_4 \end{cases}$ $V_2(x) = \begin{cases} 0 & V_1 \\ 0 & V_2 \\ -1 & V_3 \\ 1 & V_4 \end{cases}$ $u_{x_1} = V_1$, $u_{x_2} = V_2$, $V_1, V_2 \in L^P$ $(1 \le p \le \infty)$

4 $n=1$. $u \in W^{1,P}(0,1)$, $1 \le p < \infty$. (a) u is equal a.e. absolutely continuous

$u' \in L^P(0,1)$

(b) $1 < p < \infty$, $|u(x) - u(y)| \le |x-y|^{1-\frac{1}{P}} \left(\int_0^1 |u'|^P dt \right)^{\frac{1}{P}}$

Proof: (a) $u \in W^{1,P}(0,1) \Leftrightarrow \int_0^1 u\phi' dx = -\int_0^1 Du\, \phi\, dx$ $(\phi \in C_c^\infty(0,1))$, $Du \in L^P$

$V(x) = \int_0^x Du(t)\, dt$, $x \in (0,1)$ is absolutely continuous. $V' = Du$, $u = V + C$, $u' = V' = Du$

(b) $|u(x) - u(y)| = \left| \int_x^y u'(t)\, dt \right| = \left| \int_0^1 \chi(t) u'(t)\, dt \right| \le \int_0^1 |\chi(t) u'(t)| dt$ $\chi(t) = \begin{cases} 1 & x \le t \le y \\ 0 & \text{other} \end{cases}$

$$\le \left(\int_0^1 |\chi(t)|^{\frac{P}{P-1}} dt \right)^{\frac{P-1}{P}} \left(\int_0^1 |u'(t)|^P dt \right)^{\frac{1}{P}}$$

$$= |x-y|^{\frac{P-1}{P}} \left(\int_0^1 |u'(t)|^P dt \right)^{\frac{1}{P}}$$

5. U, V be open set. $V \subset\subset U$. Show exist a smooth $\zeta \equiv 1$ on V, $\zeta = 0$ near ∂U.

$V \subset\subset W \subset\subset U$. $\zeta(x) = \begin{cases} \chi_W^\varepsilon(x) & x \in V \\ 0 & x \in \mathbb{R}^n - U \end{cases}$

7. U is bdd and $\exists\ \alpha$ s.t $\alpha \cdot \nu \ge 1 \in \partial U$. $1 \le p < \infty$, $\int_{\partial U} |u|^P ds \le C \int_U |Du|^P + |u|^P dx$

Proof: $\int_{\partial U} |u|^P ds \le \int_{\partial U} |u|^P \alpha \cdot \nu\, ds = \int_U \operatorname{div}(|u|^P \alpha)\, dx = \int_U D(|u|^P) \cdot \alpha + |u|^P \operatorname{div}\alpha\, dx = \int_U P|u|^{P-1} D|u| \cdot \alpha + |u|^P \operatorname{div}\alpha\, dx$

$$\le \int_U P|u|^{P-1} |Du| + |u|^P \operatorname{div}\alpha\, dx$$

$\frac{1}{P} + \frac{1}{\frac{P}{P-1}} = 1$ $ab \le \frac{a^P}{P} + \frac{b^{\frac{P}{P-1}}}{\frac{P}{P-1}}$ $\le \int_U P\left(\frac{|Du|^P}{P} + \frac{|u|^{P-1 \cdot \frac{P}{P-1}}}{\frac{P}{P-1}} \right) + |u|^P \operatorname{div}\alpha\, dx$

$$= \int_U |Du|^P + (P-1)|u|^P dx + \int_U |u|^P \operatorname{div}\alpha\, dx$$

8. U be bdd. C^1 boundary. Prove not exist a bound linear operator $T: L^P(U) \to L^P(\partial U)$ $Tu = u|_{\partial U}$

$u \in C(\bar{U}) \cap L^P(U)$

Proof: $u_n(x) = \max\{0, 1 - n \operatorname{dis}(x, \partial U)\}$ $1 \le n < \infty$ $u_n \in C(\bar{U}) \cap L^P(U)$

$\|u_n\|_{L^P(U)} = \left(\int_U |u_n|^P dx \right)^{\frac{1}{P}} \le \left(\int_{U - U_{\frac{1}{n}}} dx \right)^{\frac{1}{P}} \to 0$ $\|u_n|_{\partial U}\|_{L^P(\partial U)} = \left(\int_{\partial U} ds \right)^{\frac{1}{P}} > 0$, not exist C

$\|u_n\|_{L^p(\partial U)} \le C\|u_n\|_{L^p(U)}$

9. Prove $\|Du\|_{L^2} \le C\|u\|_{L^2}^{\frac{1}{2}} \|D^2u\|_{L^2}^{\frac{1}{2}}$ $\quad u \in H^2(U) \cap H_0^1(U)$

proof: $\|Du\|_{L^2}^2 = \int_U |Du|^2 dx = \sum_{i=1}^n \int_U (\partial_i u)^2 dx = -\sum_{i=1}^n \int_U u \cdot \partial_{ii} u \, dx = -\int_U u \cdot \Delta u \, dx \le \int_U |u||\Delta u| dx$

$\le C\int_U |u||D^2u| \, dx \le C\|u\|_{L^2} \|D^2u\|_{L^2}$

10. (a) $\|Du\|_{L^p} \le C\|u\|_{L^p}^{\frac{1}{2}} \|D^2u\|_{L^p}^{\frac{1}{2}}$ $\quad (2 \le p < \infty)$ $\quad u \in C_c^\infty(U)$

(b) $\|Du\|_{L^{2p}} \le C\|u\|_{L^\infty}^{\frac{1}{2}} \|D^2u\|_{L^p}^{\frac{1}{2}}$ $\quad (1 \le p < \infty)$ $\quad u \in C_c^\infty(U)$

proof (a) $\|Du\|_p^p = \int_U |Du|^p dx = \int_U |Du|^{p-2}|Du|^2 dx = \sum_{i=1}^n \int_U \partial_i u (\partial_i u \, |Du|^{p-2}) \, dx$

$= -\sum_{i=1}^n \int_U u (\partial_i(\partial_i u \cdot |Du|^{p-2}) dx = \underbrace{-\int_U u \cdot \Delta u \cdot |Du|^{p-2} dx}_{I_1} \underbrace{- \sum_{i=1}^n \int_U u \cdot \partial_i u \cdot \partial_i |Du|^{p-2} dx}_{I_2}$

$I_1 = -\int_U u \cdot \Delta u \cdot |Du|^{p-2} dx \le C\int_U |u||D^2u||Du|^{p-2} dx \le C\|u\|_p \|D^2u\|_p \|Du\|_p^{p-2}$ $\quad \frac{1}{p} + \frac{}{p}$

11. U is connected $u \in W^{1,p}(U)$, $Du = 0$ in U. Prove $u \equiv C$ a.e. in U

proof: $u^\varepsilon = \eta^\varepsilon * u$ $\quad V \subset\subset U$ $\quad \varepsilon$充分小时. $Du^\varepsilon = \eta^\varepsilon * Du = (Du)^\varepsilon$ in V

$u^\varepsilon \in C^\infty(V)$, $\exists C_\varepsilon$ s.t. $u^\varepsilon = C_\varepsilon$ in V. $\quad \|u^\varepsilon\|_{L^p} = \|\eta^\varepsilon * u\|_{L^p} \le$

$\|\eta^*\|_{L^1} \|u\|_{L^p} = \|u\|_{L^p} < \infty$ $\qquad C_\varepsilon \to C \in \mathbb{R}$

$u^\varepsilon = C_\varepsilon \longrightarrow C = u$ $\qquad \|u^\varepsilon - C\|_{L^p} \to 0$ $\qquad \|u^\varepsilon - u\|_{L^p} \to 0$

12. Show example $\|D^h u\|_{L^1(V)} \le C$ for all $0 < |h| < \frac{1}{2}\mathrm{dist}(V, \partial U) \not\Rightarrow u \in W^{1,1}(V)$

proof: $U = (-0.001, 1.001)^n$ $\qquad V = (0,1)^n$ $\qquad u(x) = \begin{cases} 1 & 0 < x_1 < \frac{1}{2} \\ 0 & \text{other} \end{cases}$

$u(x) \in L^\infty(V)$ $\qquad \|D^h u\|_{L^1(V)} = \int_V |D^h u| \, dx \le \int_{\frac{1}{2}-h}^{\frac{1}{2}} \int_0^1 \cdots \int_0^1 \left|\frac{1}{h}\right| dx_n \cdots dx_1$

设 $\partial_{x_1} u$ 为弱偏导 $\quad \exists \, \partial_{x_i} u \in L^1(V)$ $\qquad = 1$

$\quad 0$

$\quad \|$

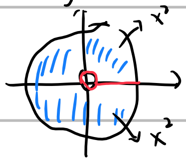
但 $u \notin W^{1,1}(V)$ $\qquad \int_V \partial_{x_1} u \, \phi \, dx = -\int_V u \, \partial_{x_1}\phi \, dx = -\int_{V \cap \{x_1 > \frac{1}{2}\}} \partial_{x_1} \phi \, dx$

13. Give an example $U \subset \mathbb{R}^n$ $\quad u \in W^{1,\infty}(U) \not\Rightarrow u$ is Lipschitz

Proof.

$U = B^0(0,1) - \{(x,y)\in B^0(0,1) \mid x \geq 0, y=0\}$ $\qquad u(x) = \{\max\{0,x\}^2 \cdot \max\{\operatorname{sgn} y, 0\}\}$

u在 U 中可微



$u(1,-1) = 1^2 \cdot 0 = 0$

$\partial_{x_1} u = 2\max\{0,x\}\max\{\operatorname{sgn} y, 0\}$ $\qquad \partial_{x_2} u = 0$ $\qquad u \in W^{1,\infty}$

$\forall \varepsilon > 0$

$$\frac{u(\frac{1}{2},\varepsilon) - u(\frac{1}{2},-\varepsilon)}{2\varepsilon} = \frac{\frac{1}{4} - 0}{2\varepsilon} \to +\infty$$

14. if $n>1$. $\quad u = \ln\ln(1+\frac{1}{|x|}) \in W^{1,n}$, $\notin L^\infty$, $\quad U = B^0(0,1)$

proof: $\partial_i u(x) = \frac{1}{\ln(1+\frac{1}{|x|})} \cdot \frac{1}{1+\frac{1}{|x|}} \cdot \left(-\frac{1}{|x|^2} \cdot \operatorname{sgn} x_i\right) = -\frac{1}{\ln(1+\frac{1}{|x|})} \cdot \frac{x_i}{|x|^2} \cdot \frac{1}{|x|+1}$

$|Du| \leq C \frac{1}{|x|} \frac{1}{\ln(1+\frac{1}{|x|})}$ $\qquad \int_{B(0,1)} |Du|^n dx \leq C \int_0^1 \left|\frac{1}{\ln(1+\frac{1}{\rho})}\right|^n \cdot \frac{1}{\rho^n} \cdot \rho^{n-1} d\rho$

$\underline{\underline{z = \ln(1+\frac{1}{\rho})}}$ $\quad C \int_{\ln 2}^{\infty} \frac{1}{z^n} dz < +\infty$

15. $\alpha > 0$. $\quad U = B^0(0,1)$ $\qquad \int_U u^2 dx \leq C \int_U |Du|^2 dx$ $\qquad |\{x \in U \mid u(x) = 0\}| \geq \alpha$

proof: $\int_U u^2 dx = \int_{U-A} (u - \langle u \rangle + \langle u \rangle)^2 dx$ $\qquad \langle u \rangle = ⨍_U u\, dx$ $\quad A = \{u(x) = 0 \mid x\}$

$$= \int_{U-A} (u - \langle u \rangle)^2 dx + 2 \underbrace{\int_{U-A} (u - \langle u \rangle) dx}_{=0} \cdot \langle u \rangle + \int_{U-A} \langle u \rangle^2 dx$$

$$\leq C \|Du\|_{L^2}^2 + \int_{U-A} \langle u \rangle^2 dx = C\|Du\|_{L^2}^2 + \langle u \rangle^2 |U-A|$$

<1

$$= C\|Du\|_{L^2}^2 + \frac{1}{|U|^2}\left|\int_{U-A} |u| dx\right|^2 \cdot |U-A| \leq C\|Du\|_{L}^2 + \frac{|U-A|^2}{|U|^2} \int_{U-A} |u|^2 dx$$

$$\|u\|_{L^2}^2 \leq C \|Du\|_{L^2}^2$$

16. (Hardy's inequality) $n \geq 3$. $\quad \int_{\mathbb{R}^n} \frac{u^2}{|x|^2} dx \leq C \int_{\mathbb{R}^n} |Du|^2 dx$

proof: 先证 $u \in C_c^\infty(\mathbb{R}^n)$ $\qquad F(x) = \frac{x}{|x|^2}$ $\qquad \int_{\mathbb{R}^n} u^2 \operatorname{div} F dx = -\int_{\mathbb{R}^n} D(u^2) \cdot F(x) dx = -2\int_{\mathbb{R}} u D(u) \cdot F(x) dx$

$\Rightarrow \left|\int_{\mathbb{R}^n} u^2 \operatorname{div} F dx\right| = 2\left|\int_{\mathbb{R}^n} Du \cdot uF dx\right| \leq 2\|Du\|_{L^2} \|uF\|_{L^2}$

$\operatorname{div} F(x) = \frac{n-2}{|x|^2}$ $\qquad |F(x)|^2 = \frac{1}{|x|^2} \Longrightarrow \frac{(n-2)^2}{4}\left(\int_{\mathbb{R}^n} \frac{u^2}{|x|^2} dx\right)^2 \leq \int_{\mathbb{R}^n} |Du|^2 dx \cdot \int_{\mathbb{R}^n} \frac{u^2}{|x|^2} dx$

$$\int_{\mathbb{R}^n} u^2 \frac{n-2}{|x|^2} dx \leq 2\left(\int_{\mathbb{R}^n} |Du|^2 dx\right)^{\frac{1}{2}} \cdot \left(\int_{\mathbb{R}^n} u^2 \frac{1}{|x|^2} dx\right)^{\frac{1}{2}}$$

$\Rightarrow \quad \frac{(n-2)^2}{4}\int_{\mathbb{R}^n}\frac{u^2}{|x|^2}dx \le \int_{\mathbb{R}^n}|Du|^2dx$

对于一般 $u\in H^1(\mathbb{R}^n)$ $\quad H^1(\mathbb{R}^n)=H_0^1(\mathbb{R}^n)$

$\exists\, u_k\in C_c^\infty(\mathbb{R}^n)$ $\quad u_k\to u$ in $H^1(\mathbb{R}^n)$ $\quad \int_{\mathbb{R}^n}|Du_k|^2dx \longrightarrow \int_{\mathbb{R}^n}|Du|^2dx$

$\frac{(n-2)^2}{4}\int_{\mathbb{R}^n}\frac{u_k^2}{|x|^2}dx$ $\quad \frac{u_k}{|x|}\in L^2(\mathbb{R}^n)$ $\quad u_k\to u$ in L^2

17. $F:\mathbb{R}\to\mathbb{R}$ C^1, F' bdd. U bdd. $u\in W^{1,p}(U)$ $1\le p\le\infty$

① $v:F(u)\in W^{1,p}$ ② $v_{x_i}=F'(u)u_{x_i}$

proof: (1) $1\le p<\infty$ $\quad u_m\in C^\infty\cap W^{1,p}$ $\quad u_m\to u$ in $W^{1,p}(U)$. $u_m\to u$ a.e.

$|F(u)-F(0)|\le \|F'\|_{L^\infty}|u| \in L^p$

18. $1\le p\le\infty$ U is bdd. ① $u\in W^{1,p}(U)\Rightarrow |u|\in W^{1,p}(U)$

② Prove $u\in W^{1,p}$ $\quad u^+,u^-\in W^{1,p}$ $\quad D^+u=\begin{cases}Du & u>0\\ 0 & u\le 0\end{cases}$ $\quad D^-u=\begin{cases}0 & u\ge 0\\ -Du & u<0\end{cases}$

$Du=D^+u-D^-u$ $\quad F_\varepsilon(r)=(\sqrt{r^2+\varepsilon^2}-\varepsilon)\chi_{\{r\ge 0\}}\in C^1(\mathbb{R})$

$-\int_U F_\varepsilon'(u)\partial_i u\,\phi\, dx=\int_U F_\varepsilon(u)\partial_i\phi\, dx$

19 $\quad Du=0$ a.e. on $\{u=0\}$

proof: ϕ C^∞ 有界. 不减 ϕ' 有界. $\phi(z)=z$ $\forall |z|\le 1$, $u^\varepsilon(x)=\varepsilon\phi(\frac{u}{\varepsilon})$

① claim $u^\varepsilon\rightharpoonup 0$ in L^2 $\quad \int_U u^\varepsilon\psi\, dx=\varepsilon\int_U\phi(\frac{x}{\varepsilon})\psi\, dx$

$\le \varepsilon\|\phi\|_{L^\infty}\|\psi\|_{L^1}\longrightarrow 0$ as $\varepsilon\to 0^+$

$\langle\psi,u^\varepsilon\rangle\longrightarrow 0$ $\quad\forall\psi$ $\quad |\phi(x)|\le\|\phi'\|_{L^\infty}|x|$ $\quad(\phi(0)=0)$

$\|u^\varepsilon\|_{L^2}^2=\varepsilon^2\int_U|\phi(\frac{x}{\varepsilon})|^2dx\le\varepsilon^2\int_U\|\phi'\|_{L^\infty}^2|\frac{u}{\varepsilon}|^2dx\le C\|u\|_{L^2}^2<\infty$

$\{\|u^\varepsilon\|_{L^2}$ 一致有界 $\quad u^\varepsilon\rightharpoonup 0$ in L^2

② $\partial_i u^\varepsilon \rightharpoonup 0$ in L^2 $\int Du^\varepsilon \cdot Du\, dx = \sum_{i=1}^n \int \partial_i u^\varepsilon \partial_i u\, dx \to 0$

$\int Du^\varepsilon \cdot Du = \sum_{i=1}^n \int \partial_i u^\varepsilon \partial_i u = \sum_{i=1}^n \int \partial_i u\, \phi'(\frac{u}{\varepsilon}) \partial_i u\, dx = \int |Du|^2 \phi'(\frac{u}{\varepsilon})\, dx$

$\{u=0\}$ $\varepsilon \to 0^+$, $Du=0$ a.e. on $\{u=0\}$

20 $u \in H^s(\mathbb{R}^n)$ $s > \frac{n}{2}$, then $u \in L^\infty$ $\|u\|_{L^\infty(\mathbb{R}^n)} \le C \|u\|_{H^s(\mathbb{R}^n)}$

chaim $\mathcal{S}(\mathbb{R}^d)$ 在 $H^s(\mathbb{R}^n)$ 中稠密 对 $\mathcal{S}(\mathbb{R}^d)$ 证明即可

$|u(x)| = |(\hat{u}(\xi))^\vee| = |\int_{\mathbb{R}^d} \hat{u}(\xi) e^{2\pi i x\cdot\xi} d\xi| \le \int_{\mathbb{R}^d} |\hat{u}(\xi)|\, d\xi = \int_{\mathbb{R}^d} |\hat{u}(\xi)| \frac{\sqrt{1+|\xi|^2}^s}{\sqrt{1+|\xi|^2}^s} d\xi$

$\le \|\frac{1}{\langle\xi\rangle^s}\|_{L^2} \|\langle\xi\rangle^s \hat{u}\|_{L^2} = C_{(s,d)} \|u\|_{H^s(\mathbb{R}^d)}$

5.21 $u, v \in H^s(\mathbb{R}^d)$ $s > \frac{d}{2}$ 则 $uv \in H^s(\mathbb{R}^d)$

proof: $\|uv\|_{H^s(\mathbb{R}^d)} = \|\widehat{uv} \langle\xi\rangle^s\|_{L^2} = \|(\hat{u} * \hat{v})\langle\xi\rangle^s\|_{L^2} = \|\int \hat{u}(\xi-\eta)\hat{v}(\eta)\langle\xi\rangle^s d\eta\|_{L^2}$

$\langle\xi\rangle^s \le C(\langle\xi-\eta\rangle^s + \langle\eta\rangle^s)$

$\|uv\|_{H^s} \le \|\int \hat{u}(\xi-\eta)\hat{v}(\eta)\langle\xi-\eta\rangle^s d\eta\|_{L^2} + \|\hat{u}(\xi-\eta)\hat{v}(\eta)\langle\eta\rangle^s d\eta\|_{L^2}$

$= \|(\hat{u}\langle\cdot\rangle^s * \hat{v})\|_{L^2} + \|\hat{u} * (\langle\cdot\rangle^s \hat{v})\|_{L^2}$

$\le \|\hat{u}\langle\xi\rangle^s\|_{L^2} \|\hat{v}\|_{L^1} + \|\hat{v}\langle\xi\rangle^s\|_{L^2} \|\hat{u}\|_{L^1}$
$= \|u\|_{H^s} \|\hat{v}\langle\xi\rangle^s \langle\xi\rangle^{-s}\|_{L^1} + \|v\|_{H^s} \|\hat{u}\langle\xi\rangle^s\langle\xi\rangle^{-s}\|_{L^1}$

$\le \|u\|_{H^s} \|\hat{v}\langle\xi\rangle^s\|_{L^2} \|\langle\xi\rangle^{-s}\|_{L^2} + \|v\|_{H^s} \|\hat{u}\langle\xi\rangle^s\|_{L^2} \|\langle\xi\rangle^{-s}\|_{L^2}$

$\le C \|u\|_{H^s} \|v\|_{H^s}$

Chapter 6 coefficients smooth and satisfy uniform elliptic condition.

1. Laplace's equation with potential function c $-\Delta u + cu = 0$ and $-\mathrm{div}(aDv) = 0$

(a) show if u solves (*) and $w > 0$ also solve (*) then $v := \frac{u}{w}$ solve (**)

for $a := w^2$ (2) v solves (**) then $u := va^{\frac{1}{2}}$ solves (*)

proof: $v=\frac{u}{w}$ $\quad v_{x_i}=\frac{u_{x_i}w-uw_{x_i}}{w^2}$ $\quad (av_{x_i})_{x_i}=(u_{x_i}w-uw_{x_i})_i = u_{x_ix_i}w+u_{x_i}w_{x_i}-u_{x_i}w_{x_i}-uw_{x_ix_i}$ $= u_{x_ix_i}w - uw_{x_ix_i}$

$-\mathrm{div}(aDv)=-(\Delta u)w+u\Delta w = -cuw+ucw = 0$

(b) $0=\mathrm{div}(aDv)=Da\cdot Dv+a\,\mathrm{div}(Dv)=Da\cdot Dv+a\Delta v$

$\Delta u=\Delta(va^{\frac12})=(\Delta v)a^{\frac12}+2Dv\cdot D(a^{\frac12})+v\Delta(a^{\frac12})$

$=(\Delta v)a^{\frac12}+a^{-\frac12}Dv\cdot Da+v\Delta(a^{\frac12})$ $\quad u=va^{\frac12}$, $v=ua^{-\frac12}$

$=(\Delta v)a^{\frac12}-a^{\frac12}\Delta v+v\Delta(a^{\frac12})=v\Delta(a^{\frac12})=\underbrace{ua^{-\frac12}\Delta(a^{\frac12})}_{c}$ //

2. $Lu=-\sum_{i,j=1}^{n}(a^{ij}u_{x_i})_{x_j}+cu$ $\quad \exists\mu.\ c(x)\geq-\mu$. bi-linear

proof: $=-\mathrm{div}(a\nabla u)+cu$

$B[u,v]=\int_U a\nabla u\cdot\nabla v+cu\cdot v\,dx$ for $u.v\in H_0^1(\Omega)$

有界性

$|B[u,v]|\leq \int_U|a\nabla u\cdot\nabla v|\,dx+\int_U|c\,u\cdot v|\,dx\leq \|a\|_{L^\infty}\|Du\|_{L^2}\|Dv\|_{L^2}+\|c\|_{L^\infty}\|u\|_{L^2}\|v\|_{L^2}$

$\leq C\|u\|_{H_0^1(\Omega)}\|v\|_{H_0^1(\Omega)}$

强制: $\theta>0$, $\quad \theta\|Du\|_{L^2}^2=\theta\int_U|Du|^2dx\leq\int_U a\nabla u\nabla u\,dx=B[u,u]-\int_U cu^2dx$

$\leq B[u,u]+\mu\|u\|_{L^2}^2\leq B[u,u]+C\mu\|Du\|_{L^2}^2$

$\mu=\frac{\theta}{2C}\Rightarrow$ $\quad \theta\|Du\|_{L^2}^2\leq B[u,u]+\frac{\theta}{2}\|Du\|_{L^2}^2$

$\|u\|_{H_0^1}^2=\|u\|_{L^2}^2+\|Du\|_{L^2}^2\leq(C+1)\|Du\|_{L^2}^2\leq\frac{2(C+1)}{\theta}B[u,u]$

3. $u\in H_0^2(U)$ is weak solution $\begin{cases}\Delta^2u=f & \text{in } U\\ u=\frac{\partial u}{\partial\nu}=0 & \text{on }\partial\Omega\end{cases}$

$\int_U\Delta u\Delta v\,dx=\int_U fv\,dx$. prove $\exists$ a unique solution.

proof: $B[u,v]=\int_U\Delta u\Delta v\,dx$ ① 有界性 $\int_U|\Delta u|\cdot|\Delta v|\,dx\leq C\|D^2u\|_{L^2}\|D^2v\|_{L^2}$

对 C_c^∞ $\quad\leq C'\|u\|_{H_0^2}\|v\|_{H_0^2}$

② $B[u,u] = \int_U \Delta u \Delta u \, dx = \sum_{j,k=1}^{n} \int_U \partial_{jj} u \, \partial_{kk} u \, dx = -\sum_{j,k}^{n} \int_U \partial_j u \, \partial_{jkk} u \, dx$

$= -\sum_{j,k}^{n} \int_U \partial_j u \, \partial_k \partial_{jk} u \, dx = \sum_{j,k}^{n} \int_U (\partial_{jk} u)^2 \, dx = \|D^2 u\|_{L^2}^2$

$\geq C(\|D^2 u\|_{L^2}^2 + \|Du\|_{L^2}^2 + \|u\|_{L^2}^2) = C\|u\|_{H_0^2(U)}^2$

对 H_0^2 $\exists \{u_m\} \subset C_c^\infty$ $\quad \|u_m - u\|_{H_0^2(\Omega)} \to 0 \quad \|u_m\|_{H_0^2(U)} \to \|u\|_{H_0^2}$

$\left| \|\Delta u_m\|_{L^2}^2 - \|\Delta u\|_{L^2}^2 \right| \leq C\|D^2(u_m - u)\|_{L^2}^2 \to 0 \quad B[u,u] \geq C\|u\|_{H_0^2}^2$

4. $u \in H^1(U)$ of $\begin{cases} -\Delta u = f & \text{in } U \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial U \end{cases}$ if $\int_U Du \cdot Dv = \int_U f v$ Prove

weak solution if and only if $\int_U f \, dx = 0$.

$\Rightarrow$ $v = 1$, $\int_U Du \cdot 0 \, dx = \int f \, dx = 0$

$\Leftarrow$ $B[u,v] = \int_U \nabla u \cdot \nabla v \, dx$ $\qquad \{u \in H^1(U) \mid \int u \, dx = 0\} = A$ $\quad C\|u\|_{H^1}^2 = \ldots + C\|Du\|_{L^2}^2$

$u \in A \quad B[u,u] = \|Du\|_{L^2}^2 = \frac{1}{2}\|Du\|_{L^2}^2 + \frac{1}{2}\|Du\|_{L^2}^2 \geq C\|u - (u)_U\|_{L^2}^2$

$\forall f \in L^2 \quad \int_U f = 0 \quad \exists! \, u_f \in H_0^1(U) \quad \int_U \nabla u_f \cdot \nabla v \, dx = \int f v$

$(f,v) = (f, v - v(v))$

$\exists u \in A$ 使 $v \in A$ 有 $B[u,v] = (f,v)$ $\quad v \in H^1 \quad v - (v)_U \in A$, $B[u, v - (v)_U]$

$B[u, \cancel{v-(v)}_U) = \int_U \nabla u \cdot \nabla(v - \nabla(v)_U) \, dx = B[u,v)$.

5. $\begin{cases} -\Delta u = f & \text{in } U \\ u + \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial U \end{cases}$

$B[u,v] = \int_U \nabla u \cdot \nabla v \, dx + \int_{\partial U} Tr u \cdot Tr v \, d = (f,v) = \int_U f v \, dx$.

check Lax–Milgram

① $|B[u,v]| \leq C\|u\|_{H^1}\|v\|_{H^1}$ (trace theorm)

② $B(u,u) = \int_U \nabla u \cdot \nabla u \, dx + \int_{\partial U} u \cdot u \, ds \geq \beta \|u\|^2_{H^1(U)}$ 若不对，则

$\|u_n\|^2_{H^1} \geq n B[u_n, u_n]$ 设 $\|u_n\| = 1 \Rightarrow B[u_n, u_n] \to 0$ $\quad \|u_n\|_{L^2} \to 0$, $\|Du_n\|_{L^2} \to 0$

$u_{n_k} \rightharpoonup u \in H^1$ L^2 $\quad u = 0$ 与 $\|u\| = 1$ 矛盾

6. $\begin{cases} -\Delta u = f & \text{in } U \\ u = 0 & \text{in } \Gamma_1 \\ \frac{\partial u}{\partial \nu} = 0 & \text{in } \Gamma_2 \end{cases}$ $\quad \partial U = \Gamma_1 \cup \Gamma_2$

proof. $\int_U f v \, dx = -\int_U -\Delta u \cdot v \, dx = -\int_{\partial U} v \frac{\partial u}{\partial \nu} ds + \int_U \nabla u \cdot \nabla v \, dx$

test function $F_1 = 0$ $\quad A = \{u \in H^1 \mid u|_{\Gamma_1} = 0\} \Leftrightarrow (f, v) = (Du, Dv)$

7. $u \in H^1(\mathbb{R}^n)$ $\quad -\Delta u + c(u) = f$, $f \in L^2(\mathbb{R}^n)$ $\quad c(u) \in L$, $c: \mathbb{R} \to \mathbb{R}$ smoth $c(0) = 0$

$c' \geq 0$ $\quad$ Prove $u \in \underline{H^2(\mathbb{R}^n)}$ $\left(\|D^2 u\|_{L^2} \leq C \|f\|_{L^2} \right)$

proof: $B[u, v] = \int_{\mathbb{R}^n} Du \cdot Dv + c(u) v \, dx$ for all $u, v \in H_0^1(\mathbb{R}^n)$

For weak solution u and all $v \in H_0^1(\mathbb{R}^n)$ $\quad B[u, v] = \int_{\mathbb{R}^n} Du \cdot Dv + c(u) v \, dx = \int_{\mathbb{R}^n} f v \, dx$

Take $v = -D_k^{-h} D_k^h u \in H_0^1(\mathbb{R}^n)$ Then $B[u, v] = -\int_{\mathbb{R}^n} Du \cdot D_k^{-h} D_k^h Du + c(u) D_k^{-h} D_k^h u \, dx$

$$= \int_{\mathbb{R}^n} D_k^h Du \cdot D_k^h Du + D_k^h c(u) D_k^h u \, dx$$

$$= \underbrace{\int_{\mathbb{R}^n} |D_k^h Du|^2 dx}_{①} + \underbrace{D_k^h c(u) D_k^h u \, dx}_{②}$$

$$= -\int_{\mathbb{R}^n} f \cdot D_k^{-h} D_k^h u \, dx \quad ③$$

对于 ② $\int_{\mathbb{R}^n} D_k^h c(u) D_k^h u \, dx = \int_{\mathbb{R}^n} \frac{c(u(x+he_k)) - c(u(x))}{h} \cdot \frac{u(x+he_k) - u(x)}{h} dx$

$$= \int_{\mathbb{R}^n} c'(\xi) \left| \frac{u(x+he_k) - u(x)}{h} \right|^2 dx \geq 0$$

对于 ③ $\int_{\mathbb{R}^n} f \cdot D_k^{-h} D_k^h u \, dx \leq \|f\|_{L^2} \|D_k^{-h} D_k^h u\|_{L^2} \leq C_\varepsilon \|D_k^h Du\|^2_{L^2} + \frac{C}{\varepsilon} \|f\|^2_{L^2}$

结合 ①②③ $\quad (1 - C_\varepsilon) \|D_k^h Du\|^2_{L^2} \leq \frac{C}{\varepsilon} \|f\|^2_{L^2}$ $\quad$ 取 $\varepsilon = \frac{1}{2C} \Rightarrow \|D_k^h Du\|_{L^2} \leq C' \|f\|_{L^2}$

$$\Rightarrow \|D^2 u\|_{L^2} \leq C \|f\|_{L^2} \Rightarrow u \in H^2(\mathbb{R}^n)$$

8. u be smoth solution $Lu = -\sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} = 0$ in U $\quad A \in W^{1,\infty}$

$v = |Du|^2 + \lambda u^2$ and show $Lv \le 0$ in U if λ is large enough. Deduce

$$\|Du\|_{L^\infty(U)} \le C\left(\|Du\|_{L^\infty(\partial U)} + \|u\|_{L^\infty(\partial U)}\right)$$

proof: $Lv = -\sum_{i,j=1}^{n} a^{ij} v_{x_i x_j} = -2\sum_{i,j} a^{ij}\left(Du_{x_i}\cdot Du_{x_j} + Du\, D_{x_i x_j} + 2\lambda u_{x_i}\cdot u_{x_j} + 2\lambda u\cdot u_{x_i x_j}\right)$

$v_{x_i} = 2|Du|\cdot Du_{x_i} + 2\lambda u\cdot u_{x_i}$

$v_{x_i x_j} = 2\left(Du_{x_j}\cdot Du_{x_i} + Du\cdot Du_{x_i x_j} + 2\lambda u_{x_j}\cdot u_{x_i} + 2\lambda u\cdot u_{x_i x_j}\right)$

$$\frac{1}{2} Lv = L(|Du|^2 + \lambda u^2) = -\sum_{i,j}\left[a_{ij}(Du)_{x_i}\cdot(Du)_{x_j} + a^{ij} Du\cdot(Du)_{x_i x_j}\right.$$

$\lambda a^{ij} u_{x_i} u_{x_j} -$

$$\left. + \lambda a^{ij} u_{x_i} u_{x_j} + \lambda a^{ij} u\cdot u_{x_i x_j}\right)$$

$$\le -\theta|D^2 u|^2 + \sum_{i,j=1}^{n} Du\cdot(Da^{ij}) u_{x_i x_j} - \lambda\theta|Du|^2$$

$$\le -\theta|D^2 u|^2 + \sum_{i,j=1}^{n}\left(\varepsilon|Du|^2 + \frac{|Da^{ij}|^2(u_{x_i x_j})^2}{4\varepsilon}\right) - \lambda\theta|Du|^2 \qquad \varepsilon = \frac{|Da^{ij}|^2}{4\theta}$$

$$= \left[\left(\sum_{i,j}\frac{|Da^{ij}|^2}{4\theta} - \lambda\theta\right]|Du|^2 \le 0 \qquad \lambda \text{ is Large}$$

$Lv \le 0$ in $U \Rightarrow \quad \|v\|_{L^\infty(U)} \le \|v\|_{L^\infty(\partial U)}$

$$\|Du\|^2_{L^\infty(U)} \le \|v\|_{L^\infty(U)} \le \|v\|_{L^\infty(\partial U)} \le \|Du\|^2_{L^\infty(\partial U)} + \lambda\|u\|^2_{L^\infty(\partial U)}$$

9. Assume u is a smooth solution of $Lu = -\sum_{ij}^{n} a^{ij} u_{x_i x_j} = f$ in U $\quad u = 0$ on $\partial\Omega$

f is bdd. Fix $x^0 \in \partial U$, A barrier at x^0 is C^2 function w such that $Lw \ge 1$ in U, $w(x^0) = 0$

$w(x^0) = 0$, $w \ge 0$ on ∂U, Show if w is barrier at x^0 then $|D(x^0)| \le C\left|\frac{\partial w}{\partial \nu}(x^0)\right|$

proof: $\partial U \in C^2$, $\Rightarrow \exists B \subset U$, s.t. $x^0 \in B \cap U$, 且B在x^0与U在x^0切向量重合. ν.

$L(u - \|f\|_\infty w) \le 0$ in U, $(u - \|f\|_\infty w)(x^0) = 0$, $(u - \|f\|_\infty w)|_{\partial U} \le 0$

Hopf lemma: $\frac{\partial u}{\partial \nu}(x^0) - \|f\|_\infty \frac{\partial w}{\partial \nu}(x^0) > 0$

Def: $w \in C^2$ 为 x^0 处闸函数 (barrier) $\begin{cases} Lw \geq 1 & \text{in } U \\ w(x_0)=0 \\ w \geq 0 & \text{on } \partial\Omega \end{cases}$

对 w 用极小值原理: $\min\limits_{\bar{U}} w = \min\limits_{\partial U} = w(x_0) = 0$

$= 0 \quad V_1(x_0)$

对 $V_1 = u + w\|f\|_{L^\infty}$, $LV_1 = Lu + Lw\|f\|_{L^\infty} \geq f + \|f\|_{L^\infty} \geq 0$ $\quad \min\limits_{\bar{U}} V_1 = \min\limits_{\partial U} V_1 = \|f\|_{L^\infty} w(x^0)$

$V_2 = u - w\|f\|_{L^\infty}$, $LV_2 = Lu - Lw\|f\|_{L^\infty} \leq f - \|f\|_{L^\infty} \leq 0$, $\quad \max\limits_{\bar{U}} V_2 = \max\limits_{\partial U} V_2 = V_2(x_0)$

$0 \geq \frac{\partial V_1}{\partial \nu}(x_0) = \frac{\partial u}{\partial \nu}(x_0) + \|f\|_{L^\infty}\frac{\partial w}{\partial \nu}(x_0) \qquad 0 \leq \frac{\partial V_2}{\partial \nu}(x_0) = \frac{\partial u}{\partial \nu}(x_0) - \|f\|_{L^\infty}\frac{\partial w}{\partial \nu}(x^0)$

$u = 0$ on $\partial U \Rightarrow \quad |\nabla u(x^0)| = \left|\frac{\partial u}{\partial \nu}(x^0)\right| \leq \|f\|_{L^\infty}\left|\frac{\partial w}{\partial \nu}(x^0)\right|$

10. $\begin{cases} -\Delta u = 0 & \text{in } U \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial U \end{cases}$ show $u \equiv C$ for some C

proof: ① $\underbrace{\int_U -\Delta u \cdot u}_{0} = \int_U |\nabla u|^2 dx + \int_{\partial U} u\frac{\partial u}{\partial \nu} dS = \int_U |\nabla u|^2 dx$

$\Rightarrow |\nabla u|^2 = 0 \overset{a.e.}{\Rightarrow} u \equiv C$ a.e. $\Rightarrow u \in C^\infty(\bar{U}) \Rightarrow u \equiv C$

② 若 u 在 U 内部取得极大值,则 $u \equiv C$, 若 $x^0 \in \partial U$. $u(x^0) = \sup\limits_{\bar{U}} u(x)$, $u(x^0) > u(x)$, 则 Hopf $\frac{\partial u}{\partial \nu}(x^0) > 0$ 矛盾

11. $u \in H^1(U)$ is bdd. weak solution of $-\sum\limits_{i,j}(a^{ij}u_{x_i})_{x_j} = 0$ in U, let $\phi: \mathbb{R} \to \mathbb{R}$ be convex and smooth $w = \phi(u)$. Show w is weak solution subsolution. ($B[w,v] \leq 0$) all $v \geq 0$ $\in H_0^1$

proof: $B[w,v] = \int_U \sum\limits_{i,j=1}^n a^{ij}[\phi(u)]_{x_i} v_{x_j} dx = \int_U \sum\limits_{i,j=1}^n a^{ij}\phi'(u)u_{x_i}v_{x_j} dx$

$= \int_U \sum\limits_{i,j=1}^n a^{ij}u_{x_i}(\phi'(u)v)_{x_j} - \sum\limits_{i,j=1}^n a^{ij}u_{x_i}u_{x_j}\phi''(u)v\, dx$

$\leq 0 - \theta\int_U |Du|^2 \underbrace{\phi''(u)}_{>0} v\, dx \leq 0$

12. $Lu = -\sum\limits_{i,j=1}^n a^{ij}u_{x_ix_j} + \sum\limits_{i=1}^n b^i u_{x_i} + cu$ satisfy weak maximum principle

$\exists v \in C^2(U) \cap C(\bar{U})$ $\begin{cases} Lv \geq 0 & \text{in } U \\ v > 0 & \text{on } \partial U \end{cases}$, 求证: $\forall u \in C^2(U)$ $\begin{cases} Lu \leq 0 & \text{in } U \\ u \leq 0 & \text{on } \partial U \end{cases} \Rightarrow u \leq 0$ in U

proof: $u \in C^2(U) \cap C(\bar{U})$. $Lu \leq 0$ in U $\quad u \leq 0$ on ∂U, $w = \frac{u}{v} \in C^2(U) \cap C(\bar{U})$

Def Mw. $w_{x_i} = \left(\frac{u}{v}\right)_{x_i} = \frac{u_{x_i}v - uv_{x_i}}{v^2} \Rightarrow (v^2 w_{x_i})_{x_j} = (u_{x_i}v - uv_{x_i})_{x_j}$

$= u_{x_ix_i}v + u_{x_i}v_{x_j} - u_{x_j}v_{x_i} - uv_{x_ix_j}$, $-\sum_{i,j=1}^{n} a^{ij}(v^2 w_{x_i})_{x_j} = -v\sum_{i,j=1}^{n} a^{ij}u_{x_ix_j} + u\sum_{i,j=1}^{n} a^{ij}v_{x_ix_j}$

$= vLu - uLv - v^2\sum_{i=1}^{n} b^i w_{x_i} = Mw - v^2\sum_{i=1}^{n} b^i w_{x_i}$, $Mw = \underset{>0}{\underline{v}}\,\underset{<0}{\underline{Lu}} - \underset{>0}{\underline{u}}\,\underset{>0}{\underline{Lv}} \le 0$ in $\{u>0\}$

M is uniform elliptic $0 < \max_{\{u>0\}} \frac{u}{v} = \max_{\{u>0\}} w = \max_{\partial\{u>0\}} w = 0$

if $\{u>0\} \neq \emptyset \Rightarrow \{u>0\} = \emptyset$. $u \le 0$ in U

13. $L = -\sum_{i,j=1}^{n} (a^{ij}u_{x_i})_{x_j}$, operator L, with zero boundary, has eigenvalues

$0 < \lambda_1 < \lambda_2 \le \cdots$, $\lambda_k = \max_{S\in\Sigma_{k-1}} \min_{\substack{u\in S^\perp \\ \|u\|_{L^2}=1}} B[u,u]$, Σ_{k-1} $(k-1)$ d subspace of $H_0^1(U)$

Proof: 作 $A = L^{-1}: L^2 \longrightarrow H_0^1(U) \hookrightarrow L^2(U)$, A为 $L^2(U) \to L^2(U)$ 紧算子

$Lu=f$, $L^{-1}f = u$ $f \to u \longrightarrow u$ 设λ_k特征函数为w_k , $\langle w_i, w_j\rangle_{L^2} = \delta_{ij}$

$Lw_k = \lambda_k w_k \Rightarrow Aw_k = \frac{1}{\lambda_k} w_k$ A的特征值 $\lambda_1^{-1} > \lambda_2^{-1} \ge \cdots \ge \ > 0$

$u = \sum_i (u, e_i)e_i$

(Hilbert-Schmidt theorem) $Ae_k = \lambda_k^{-1}e_k$ $\|e_k\|_{L^2(U)} = 1$ $\forall f\in L^2(U)$ $f = \sum_i (f, e_i)e_i$

$\{e_k\}_{k\in\mathbb{Z}^+}$ 为$L^2(U)$标准正交基 , $B[u,u] = \langle f, u\rangle = \langle Lu, u\rangle = \langle \sum_i (u,e_i)(\lambda_i e_i), \sum_i (u,e_i)e_i\rangle$ $\forall u$, $\|u\|_{L^2(U)} = 1$

span$\{e_1 \cdots e_k\} \in \Sigma_{k-1}$ span$\{e_{i_1} \cdots e_{i_k}\} \in \Sigma_{k-1}$ $= \sum_{i=1}^{\infty} \lambda_i \langle u, e_i\rangle^2$

Σ_{k-1}是 $H_0^1(U)$ 全体$k-1$维子空间.

① $\forall S\in\Sigma_{k-1}$, $\exists u_k \in$ span$\{e_1 \cdots e_k\}$ $\exists u_k \perp S$ (by Hilbert-Schmidt theorem)

$\Rightarrow \inf_{\substack{\|u\|_{L^2}=1 \\ u\in S^\perp}} B[u,u] \le B[u_k, u_k] = \sum_{i=1}^{k} \lambda_i \langle u_k, e_i\rangle^2 \le \lambda_1\alpha_1^2 + \lambda_2\alpha_2^2 + \cdots \lambda_k\alpha_k^2 \le \lambda_k(\alpha_1^2 + \cdots \alpha_k^2) = \lambda_k$

② $S =$ span$\{e_1 \cdots e_{k-1}\}$ 则 $\forall u\in S^\perp$ | $\lambda_k = B[e_k, e_k] \le \sum_{j\ge k} \lambda_j (u, e_j)^2 = B[u,u]$

$S =$ span$\{e_1 \cdots e_{k-1}\}$ $u_k \in$ span$\{e_k\} \subset S^\perp$

$S^\perp =$ span$\{e_k, e_{k+1} \cdots\cdots\}$

$\Rightarrow \inf_{\substack{\|u\|_{L^2}=1 \\ u\in S^\perp}} B[u,u] \le B[u_k, u_k] = \lambda_k = B[e_k, e_k] \le \sum_{j\ge k} \lambda_j (u, e_j)^2 = B[u,u]$

$\lambda_k \ge \sup_{S\in\Sigma_{k-1}} \inf_{\substack{\|u\|_{L^2}=1 \\ u\in S^\perp}} B[u,u]$ | $\le$: 取 $S =$ span$\{\frac{e_1}{\sqrt{\lambda_1}} \cdots \frac{e_k}{\sqrt{\lambda_k}}\}$ 为$H_0^1(U)$标准基 内积为 $B[\ ,\ ]$.

从而 $\forall u\in S^\perp$且 $\|u\|_{L^2} = 1$, $u = \sum_{j\ge k} (a_j\sqrt{\lambda_j})\frac{e_j}{\sqrt{\lambda_j}}$

$B[u,u] = \sum_{j\ge k} a_j^2\lambda_j \ge \lambda_k$

14. λ_1 be the principal eigenvalue of elliptic.

$Lu = -\sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} + cu$ Prove "max − min"

$$\lambda_1 = \sup_u \inf_x \frac{Lu(x)}{u(x)} \quad (u \in C^\infty(\bar{U}),\ u>0 \text{ in } U,\ u=0 \text{ on } \partial U)$$

proof: $X = \{u \in C^\infty(\bar{U}) \mid u>0 \text{ in } U,\ u=0 \text{ on } \partial U\}$, $Lw_1 = \lambda_1 w_1$, $w_1 \in X$

$\inf_{x\in U} \frac{Lw_1}{w_1} = \boxed{\lambda_1 \le \sup_{u\in X} \inf_{x\in U} \frac{Lu}{u}}$ 下证 $\lambda_1 \geq \sup_{u\in X} \inf_{x\in U} \frac{Lu}{u}$

$\forall u \in X$, $\inf_{x\in U} \frac{Lu}{u} \le \lambda_1 \Leftrightarrow \inf_{x\in U} (Lu - \lambda_1) \le 0$ consider $L^* w_1^* = \lambda_1 w_1^*$

$\Leftrightarrow \langle L^* w_1^*, u\rangle = \langle \lambda_1 w_1^*, u\rangle \Leftrightarrow \langle Lu, w_1^*\rangle = \langle \lambda_1 u, w_1^*\rangle$

$\Leftrightarrow \langle Lu - \lambda_1 u, w_1^*\rangle = 0 \Rightarrow \inf_x (Lu - \lambda_1 u) \le 0$

15. $U(\tau) \subseteq \mathbb{R}^\infty$ (关于 τ C^∞), $\partial U(\tau)$ 速度为 v. $\forall \tau$. $\begin{cases} -\Delta w = \lambda w & \text{in } U(\tau) \\ w = 0 & \text{on } \partial U(\tau) \end{cases}$

$\|w\|^2_{L^2(U(\tau))} = 1$, $\lambda = \lambda(\tau) \in C^\infty$, $w = w(x,\tau) \in C^\infty_{\tau,x}$ 利用 $\frac{d}{d\tau}\int_{U(\tau)} f\,dx = \int_{\partial U(\tau)} f(\vec{v}\cdot\nu)\,ds$

$+ \int_{U(\tau)} \partial_\tau f\,dx$ 去证明 $\overset{\left(\frac{d\lambda}{d\tau}\right)}{\overset{\shortparallel}{\dot\lambda}} = -\int_{\partial U(\tau)} \left|\frac{\partial w}{\partial \nu}\right|^2 (\vec{v}\cdot\nu)\,ds$

proof: $-\Delta w = \lambda w \Rightarrow -\int_{U(\tau)} w\cdot\Delta w = \lambda\int_{U(\tau)} w^2 = \lambda = \int_{U(\tau)} |\nabla w|^2 dx$

$\lambda = \int_{U(\tau)} |\nabla w|^2 dx$ 对 τ 求导 $\dot\lambda = \int_{\partial U(\tau)} |\nabla w|^2 (\vec{v}\cdot\nu)\,ds + \int_{U(\tau)} \partial_\tau |\nabla w|^2 dx$

由 $w=0$ on $\partial U(\tau) \Rightarrow$ $\nabla w = \frac{\partial w}{\partial \nu}$ $\dot\lambda = \int_{\partial U(\tau)} \left|\frac{\partial w}{\partial \nu}\right|^2 (\vec{v}\cdot\nu)\,ds + \underline{\int_{U(\tau)} \partial_\tau |\nabla w|^2 dx}$

第二项 $= \int_{U(\tau)} 2\nabla w\cdot\nabla(\partial_\tau w)\,dx = \int_{U(\tau)} 2w(-\Delta\partial_\tau w)\,dx = \int_{U(\tau)} 2w\cdot\partial_\tau(\lambda w)\,dx$

$= \int_{U(\tau)} 2\lambda w\cdot\partial_\tau w\,dx + \int_{U(\tau)} 2w^2\cdot\dot\lambda\,dx = 2\dot\lambda + 2\lambda\int_{U(\tau)} \partial_\tau w^2 dx = 2\dot\lambda + 2\lambda\Big[\underbrace{\frac{d}{d\tau}\int_{U(\tau)} w^2 dx}_{0} - \underbrace{\int_{\partial U(\tau)} \cdots}_{0}\Big]$

$= 2\dot\lambda$

$$\Rightarrow \dot\lambda = -\int_{\partial U(\tau)} \left|\frac{\partial w}{\partial \nu}\right|^2 (\vec{v}\cdot\nu)\,ds$$

16 (Radiation condition) $-\Delta w=\lambda w$ in $\mathbb{R}^n$ $\lambda:=\sigma^2$ (*)

(a) $w=e^{i\sigma w\cdot x}$ solves (*) $|w|=1$, $u=e^{i\sigma(w\cdot x-t)}$ sole wave equation

(b) $n=3$ $\phi:=\frac{e^{i\sigma x}}{4\pi|x|}$ solves $-\Delta\phi=\lambda\phi+\delta_0$ in $\mathbb{R}^3$

(c) Sommerfeld radiation (*) $\lim_{r\to\infty} r(w_r-i\sigma w)=0$, $w_r=Dw\cdot\frac{x}{|x|}$ (b) dues (a) not

proof: (a) $-\Delta w=-\Delta e^{i\sigma w\cdot x}=-(i\sigma w)^2 e^{i\sigma w\cdot x}=\sigma^2 e^{i\sigma w\cdot x}=\lambda w$

(b)

17: w satify radition the $w\equiv 0$

chapter 7.

1. Prove there is most one smooth solution $\begin{cases} u_t-\Delta u=f & \text{in } U_T \\ \frac{\partial u}{\partial\nu}=0 & \text{on } \partial U\times[0,T] \\ u=g & \text{on } U\times\{t=0\}\end{cases}$

proof: u_1,u_2为解. $v=u_1-u_2$ 需证 $v=0$ $\begin{cases} v_t-\Delta v=0 & \text{in } V_T \\ \frac{\partial v}{\partial\nu}=0 & \text{on } \partial U\times[0,T] \\ v=0 & \text{on } U\times\{t=0\}\end{cases}$

$\int_U v\cdot v_t-v\Delta v\,dx=0=\int_U\frac{1}{2}\frac{d}{dt}v_t^2-|\nabla v|^2dx\Rightarrow\frac{1}{2}\frac{d}{dt}\|v(t)\|_{L^2}^2=-\int_U|\nabla v|^2dx\le 0$

又 $\|v(0)\|_{L^2}^2=0\Longrightarrow\|v(t)\|_{L^2}^2\le 0\Longrightarrow v(t)=0$ in $(0,T]$

2. $\begin{cases} u_t-\Delta u=0 & \text{in } V\times(0,\infty) \\ u=0 & \text{on } \partial U\times[0,\infty) \\ u=g & \text{on } V\times\{t=0\}\end{cases}$ Prove $\|u(\cdot,t)\|_{L^2(U)}\le e^{-\lambda_1 t}\|g\|_{L^2(U)}$ $\forall t\ge 0$

$\lambda_1>0$ is the principle eigenvalue of $-\Delta$. with zero boundary.

proof: $\frac{1}{2}\frac{d}{dt}\|u(t)\|_{L^2}^2=-\int_U|\nabla u|^2dx$, $\lambda_1=\inf_{\substack{u\in H_0^1(U)\\u\neq 0}}\frac{\|Du\|_{L^2(U)}^2}{\|u\|_{L^2(U)}^2}$ $\eta'(t)\le\phi(t)\eta(t)+\psi(t)$

$\le-\lambda_1\|u(t)\|_{L^2}^2$ $\int|\nabla u|^2>\lambda_1\int u^2dx$ $\Rightarrow\eta(t)\le e^{\int_0^T\phi}(\eta(0)+\int_0^T\psi)$

Gronwall Inequality. $\Longrightarrow$ $\frac{d}{dt}\|u(t)\|_{L^2(U)}^2\le-2\lambda_1\|u(t)\|_{L^2}^2\Longrightarrow\|u(t)\|_{L^2}^2\le e^{\int_0^T-2\lambda_1}(g+)=e^{-2\lambda_1 t}\|g(t)\|_{L^2}^2$

3. $\begin{cases}\partial_t u+Lu=0 & \text{in } U_T \\ u=0 & \text{on } \partial U\times[0,T] \\ u=g & \text{on } U\times\{t=0\}\end{cases}$ $\begin{cases}\partial_t v-L^*v=0 & \text{in } V_T \\ v=0 & \text{on } \partial U\times[0,T] \\ v=h & \text{on } U\times\{t=T\}\end{cases}$ 求证 $\int_U g(x)v(x,0)dx=\int_U u(x,T)h(x)dx$

proof: $\int_U u(x,T)h(x)-g(x)v(x,0)dx=\int_U u(x,T)v(x,T)-u(x,0)v(x,0)\,dx$

$$= \int_U \int_0^T (\partial_t u)\cdot v + (\partial_t v)\cdot u \, dt \, dx = \int_0^T \int_U (\partial_t u)\cdot v + (\partial_t v)\cdot u \, dx\, dt$$

$$\langle Lu, v\rangle = \langle u, L^* v\rangle = \int_0^T \int_U (\partial_t u)\cdot v + Lu\cdot v - u\cdot L^* v + (\partial_t v)\cdot u \, dx\, dt$$

$$= \int_0^T \int_U (\partial_t u + Lu) v + u(\partial_t v - L^* v) \, dx \, dt = 0$$

4. (Galerkin's method for Possion's equation) $f \in L^2(U)$ and

$u_m = \sum_{k=1}^m d_m^k w_k$ solves $\int_U Du_m \cdot Dw_k \, dx = \int_U f \cdot w_k \, dx$ for $k = 1, 2 \cdots m$

$\{w_k\}$ is an orthogonal basis of $H_0^1(U)$. Show $\{u_m\}_{m=1}^\infty$ converge weakly

in $H_0^1(U)$ to weak solution of $\begin{cases} -\Delta u = f & \text{in } U \\ u = 0 & \text{on } \partial U \end{cases}$

Proof: step1: $\{u_m\}$ 在 H_0^1 中一致有界 $\int_U Du_m \cdot Dw_k \, dx = \int_U f \cdot w_k \, dx$ 同乘 d_m^k 且对 k 求和.

$$\Rightarrow \int_U Du_m \cdot Dw_k \cdot d_m^k dx = \int_U f \cdot w_k \cdot d_m^k \, dx \Rightarrow \int_U Du_m \sum_{k=1}^m Dw_k \cdot d_m^k \, dx = \int_U f \cdot \sum_{k=1}^m w_k d_m^k \, dx$$

$$\Rightarrow \|Du_m\|_{L^2}^2 = \int_U f u_m \, dx \le \|f\|_{L^2} \|u_m\|_{L^2} \le \varepsilon \|u_m\|_{L^2}^2 + C(\varepsilon)\|f\|_{L^2}^2 \le \varepsilon \|u_m\|_{H_0^1}^2 + C(\varepsilon)\|f\|_{L^2}^2$$

$$\Rightarrow \|u_m\|_{H_0^1} \le C \|f\|_{L^2}$$

step 2 ∃子列 $u_{m_k} \rightharpoonup u$ in $H_0^1(U)$, $Du_{m_k} \rightharpoonup v$ in $L^2(U)$

problem u is solution? v is Du?

$$\int f\varphi = \int Du_{m_k} D\varphi = -\int u_{m_k} \cdot D^2\varphi = -\int u \cdot D^2 \varphi = \int Du \cdot D\varphi$$
(with $\int v \cdot D\varphi$ written above the second equals sign: $\overset{\int v\cdot D\varphi}{=}$)

Chapter 8 (本节假设 $U \subseteq \mathbb{R}^n$ 有界开集, $\partial U \in C^\infty$)

8.1 求证① $u_k(x) = \sin(kx) \rightharpoonup 0$ in $L^2(0,1)$ 求证: $u_k \rightharpoonup \lambda a + (1-\lambda) b$ in $L^2(0,1)$

② 固定 $a, b \in \mathbb{R}$, $0 < \lambda < 1$ 定义 $u_k(x) := \begin{cases} a & \frac{j}{k} \le x < \frac{j+\lambda}{k} \\ b & \frac{j+\lambda}{k} \le x < \frac{j+1}{k} \end{cases}$ $j = 0, \cdots k-1$

Proof: ① $\forall \varphi \in L^2(0,1) \Rightarrow \varphi \in L^1(0,1)$ 则 $\langle u_k, \varphi\rangle = \int_0^1 \varphi(x) \sin kx \, dx$ by Riemann–Lebesgue

Lemma $\langle \varphi_k, \varphi\rangle \longrightarrow 0$ $k \to +\infty$ 则 $u_k \rightharpoonup 0$ in $L^2(0,1)$

② $1 \le p \le \infty$, $f \in L^p_{loc}(\mathbb{R})$ 且周期为T, 令 $u_k(x) = f(kx)\chi_{[0,1]}(x)$ 则 $u_k \rightharpoonup \frac{1}{T}\int_0^T f(x)dx$ in $L^p(0,1)$

令 $f(x)=\begin{cases} a & [j,j+\lambda) \\ b & [j+\lambda,j+1) \end{cases}$ $u_k(x)=\begin{cases} a & [\frac{j}{k},\frac{j+\lambda}{k}) \\ b & [\frac{j+\lambda}{k},\frac{j+1}{k}) \end{cases}$ 则 $u_k \longrightarrow \frac{1}{1}\int_j^{j+\lambda} a\,dx+\int_{j+\lambda}^{j+1} b\,dx$

$= \lambda a+(1-\lambda)b$

2. Let $\phi: U\to\mathbb{R}$ Find $L=L(p,z,x)$ so that PDE $-\Delta u+D\phi\cdot Du=f$'s Euler-Lagrange equation

Proof: $-\Delta \Leftrightarrow |\nabla u|^2$. $f\Leftrightarrow f(x)\cdot z$ $\nabla\phi \Leftrightarrow e^{-\varphi}$

$L(p,z,x)=e^{-\varphi(x)}\left(\frac{1}{2}|p|^2-f(x)\cdot z\right)$ $I[w]=\int_U L(\nabla w,w,x)\,dx=\int_U e^{-\varphi(x)}\left(\frac{1}{2}|\nabla w|^2-f(x)\cdot w\right)dx$

Euler-Lagrange equation $-\sum_{i=1}^n (\partial_{p_i}L(\nabla u,u,x))_{x_i}+\partial_z L(\nabla u,u,x)=0 \Longrightarrow -\sum_{i=1}^n (e^{-\varphi(x)}u_{x_i})_{x_i}-(e^{-\varphi(x)}f(x))=0$

$\Rightarrow -\sum_{i=1}^n \left((e^{-\varphi(x)})_{x_i}u_{x_i}\right)-\sum_{i=1}^n (e^{-\varphi(x)}u_{x_ix_i})-e^{-\varphi(x)}f(x)=0$

$\Rightarrow e^{-\varphi(x)}(\nabla\varphi\nabla u-\Delta u-f)=0 \Rightarrow -\Delta u+\nabla\phi\cdot\nabla u=f$ in U

8.3 The elliptic regularition of the heat equation is the PDE $(*)$

$u_t-\Delta u-\varepsilon u_{tt}=0$ in U_T, $\varepsilon>0$, 求证: $(*)$ 是某个 $I_\varepsilon[w]:=\iint_{U_T} L_\varepsilon(\nabla w,w_t,w,x,t)\,dx\,dt$ 对应 Euler-Lagrange equation.

$L(p,z,x)=\frac{1}{2}e^{-\frac{t}{\varepsilon}}(p_1^2+\cdots p_n^2+\varepsilon p_{n+1}^2)$ $I[w]:=\frac{1}{2}\iint_{U_T} e^{-\frac{t}{\varepsilon}}(|\nabla_x w|^2+\varepsilon|w_t|^2)\,dx\,dt$

4. $\eta:\mathbb{R}^n\to\mathbb{R}$ is C^1 (a) show $L(p,z,x)=\eta(z)\det P$ ($P\in M^{n\times n}$, $z\in\mathbb{R}^n$) is a null Largrange

10. existence , $\begin{cases} -\Delta u=|u|^{q-1}u & \text{in } U \\ u=0 & \text{on } \partial U \end{cases}$ $1<q<\frac{n+2}{n-2}$, $n>2$

Proof: $I[w]=\frac{1}{2}\int_U|\nabla w|^2dx-\frac{1}{q+1}\int_U|w|^{q+1}dx$ $\langle I'[u],v\rangle=\int_\Omega \nabla u\nabla v-|u|^{q-1}u\cdot v\,dx$

① $I\in C^1$ ② $I[0]=0$ ③ Palais-smale condition each $\{u_k\}$ $\begin{cases} I[u_k] \text{ bdd in } H \\ I'[u_k]\to 0 \text{ in } H \end{cases}$